HyStart++ (RFC 96) over IEEE 802.11g + IEEE 802.15.4

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1 Introduction

TCP slow start was designed to probe available bandwidth cautiously — cwnd doubles every RTT, growing exponentially until loss is detected. The fundamental problem is that by the time loss occurs, the sender has already massively overshot the bottleneck capacity, causing a burst of dropped packets and a sharp cwnd reduction. The network experiences unnecessary congestion that the sender itself created.
HyStart++ (RFC 9406, 2023) is a refined successor that introduces a middle ground between aggressive slow start and full congestion avoidance. Instead of exiting directly to CA when a delay increase is detected, it transitions to a Conservative Slow Start (CSS) phase where cwnd continues to grow, but at a fraction (1/4) of the normal rate. This cautious probing phase serves two purposes: it continues filling the pipe if the delay signal was spurious, and it gently backs off if the congestion was real. The algorithm operates on a round-based RTT measurement scheme — one round corresponds approximately to one RTT, tracked via sequence numbers rather than timers. Within each round it computes a minimum RTT and compares it to the previous round's minimum. If the increase exceeds an adaptive threshold — computed as a clamped fraction of the baseline RTT — the transition to CSS is triggered. This report presents the implementation and evaluation of HyStart++ (RFC 96) in NS-3 over two network topologies:

- IEEE 802.11g Wi-Fi (Static)
- IEEE 802.15.4 LR-WPAN (Mobile)

2 Network Topologies

2.1 IEEE 802.11g Wi-Fi (Static)

A multi-hop ad-hoc Wi-Fi mesh network with nodes placed on a uniform grid.

- Data rate: 24 Mbps
- Static grid layout (no mobility).
- Nodes placed in a square grid.
- Coverage area = coverageMult × txRange (mainly tested at coverageMult = 1×).
- MAC/PHY: AdhocWifiMac + YansWifiPhy with 802.11g (24 Mbps data rate).
- Propagation model: RangePropagationLossModel.
  Energy model: WifiRadioEnergyModel on every node
- MAC: AdhocWifiMac
- Routing: OLSR
- Transport: TCP with TcpHyStartPlusPlus (RFC 96).
- TCP CC: HyStart++

2.2 IEEE 802.15.4 LR-WPAN (Mobile)

Nodes move using Random Waypoint Mobility Model in a 200m × 200m area.

- PHY rate: 250 kbps
- MAC: CSMA/CA
- Network: IPv6 with 6LoWPAN
- Transport: UDP only

3 Parameters

3.1 Common Parameters

Parameter	Values
Nodes	20–100
Flows	10–50
Packets/sec	100–500
Simulation time	s

- speed(ms): 5.0, 10.0, 15.0, 20.0, 25.0 ms (main varying parameter)for wpan technology
- Coverage area (square coverage are varying one side as Txrange, 2 x Txrange, 3 x Txrange, 4 x Txrange, and 5 x Txrange) for the case of having static nodes only

3.2 Metrics

- Throughput
- End-to-End Delay
- Packet Delivery Ratio (PDR)
- Packet Drop Ratio
- Energy Consumption

4 Some Results from the csv file

4.1 for wifi

Nodes	Flows	PPS	Tx Range	Throughput (Mbps)	Delay (ms)	PDR (%)	Drop (%)	Energy (J)
20	10	100	2	2.8306	0.504	99.89	0.09	668.36
20	10	100	3	2.8306	0.505	99.89	0.09	668.36
20	10	100	4	4.6621	0.615	99.95	0.04	676.98
20	10	100	5	4.6621	0.615	99.95	0.04	676.98
40	30	200	2	2.2138	1.670	99.66	0.30	1342.75
40	30	200	3	2.3761	2.796	99.45	0.45	1344.15
40	30	200	4	2.3968	3.149	99.26	0.41	1345.89
40	30	200	5	2.3922	5.096	99.01	0.78	1348.64
60	10	300	4	1.6990	1.720	99.80	0.17	2016.09
60	10	300	5	1.7009	1.570	99.72	0.27	2016.54
60	10	400	3	2.3594	3.716	99.42	0.16	2025.74
100	10	300	3	0.9165	17.433	98.14	1.10	3387.14
100	10	300	4	0.8593	19.853	98.58	1.14	3389.82
100	10	300	5	0.9615	22.799	97.33	2.18	3393.83
100	10	400	2	1.3013	21.500	97.93	1.77	3396.78

Table 2: IEEE 802.11g Wi-Fi Static Network Performance (Compact View)

4.2 for wifi-mobile

Nodes	Flows	PPS	Speed (m/s)	Throughput (Mbps)	Delay (ms)	PDR (%)	Drop (%)	Energy (J)
20	10	100	5	0.1016	242.97	84.68	12.11	2.69
20	10	100	10	0.0924	268.80	81.67	15.11	2.67
20	10	100	15	0.0979	242.33	83.52	14.42	2.68
20	10	100	20	0.0690	238.21	75.36	20.13	2.58
40	20	400	15	0.0915	271.02	73.66	21.33	4.94
40	20	400	20	0.0817	310.83	72.21	23.60	4.90
40	20	400	25	0.1056	332.65	78.77	17.41	4.99
40	20	500	5	0.1030	311.66	77.37	18.93	4.98
60	30	200	25	0.0834	423.45	75.44	20.38	7.13
60	30	300	5	0.1151	355.04	76.37	18.98	7.29
60	30	300	10	0.0850	348.83	71.95	22.46	7.16
60	30	300	15	0.0935	418.26	74.10	21.79	7.19
80	40	300	20	0.0970	560.34	71.56	23.71	9.45
80	40	300	25	0.0859	486.88	75.31	19.79	9.39
80	40	400	5	0.1006	489.67	76.23	19.27	9.45
80	40	400	10	0.0872	458.22	72.64	21.88	9.41
100	40	300	5	0.0925	461.69	71.82	22.64	11.68
100	40	300	10	0.0945	488.42	74.93	20.63	11.67
100	40	300	15	0.1047	421.13	78.90	15.96	11.70

Table 3: IEEE 802.15.4 Mobile Network Performance (Compact View)

5 Modifications made in the simulator

- CP Congestion Control: Switched from default (NewReno / CUBIC) to TcpHyStartPlusPlus (TcpHyStartPlusPlus class in NS-3).
- Enabled logging for HyStart++ events (HYSTART-CSS-ROUND, HYSTART=¿PHASE, CSS → CA transitions) to verify correct operation.
- Added energy consumption tracing using WifiRadioEnergyModel.
- Custom result collection and CSV output for throughput, delay, PDR, drops, and energy.
- Grid positioning logic with variable coverageMult and node scaling.
- No major core NS-3 source code changes; all modifications were in the scratch simulation script (hystart++ algorithm then used) and helper classes.

6 HyStart++ Implementation

HyStart++ was implemented as a subclass of TcpCongestionOps.

6.1 Key Methods

- `PktsAcked()`
- `IncreaseWindow()`
- `GetSsThresh()`
- `CongestionStateSet()`

6.2 Important Constants

Constant	Value
MIN_RTT_THRESH	4 ms
MAX_RTT_THRESH	16 ms
CSS_ROUNDS	5
L	8

7 RTT Smoothing Extension

A weighted moving average was used:

$$\mathrm{RTT}_{smooth} = 0.2 \cdot RTT_{t-2} + 0.3 \cdot RTT_{t-1} + 0.5 \cdot RTT_t$$

This reduces noise from wireless jitter.

8 Results with Graphs

8.1 Wifi 802.11.4

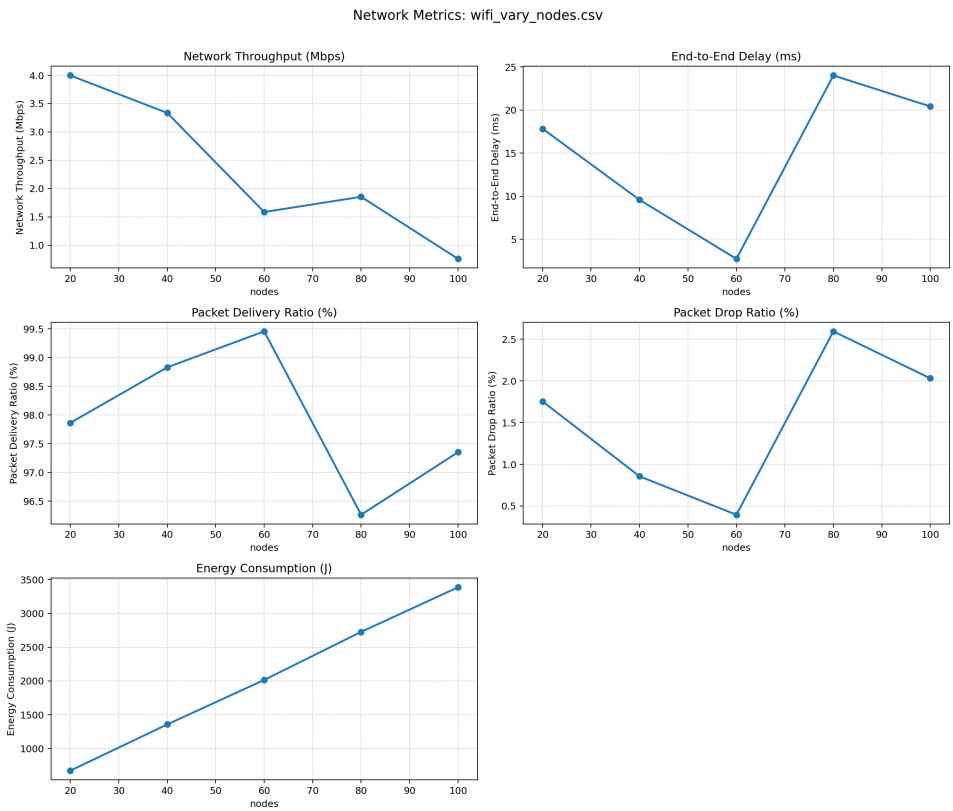

Figure 1: varied nodes

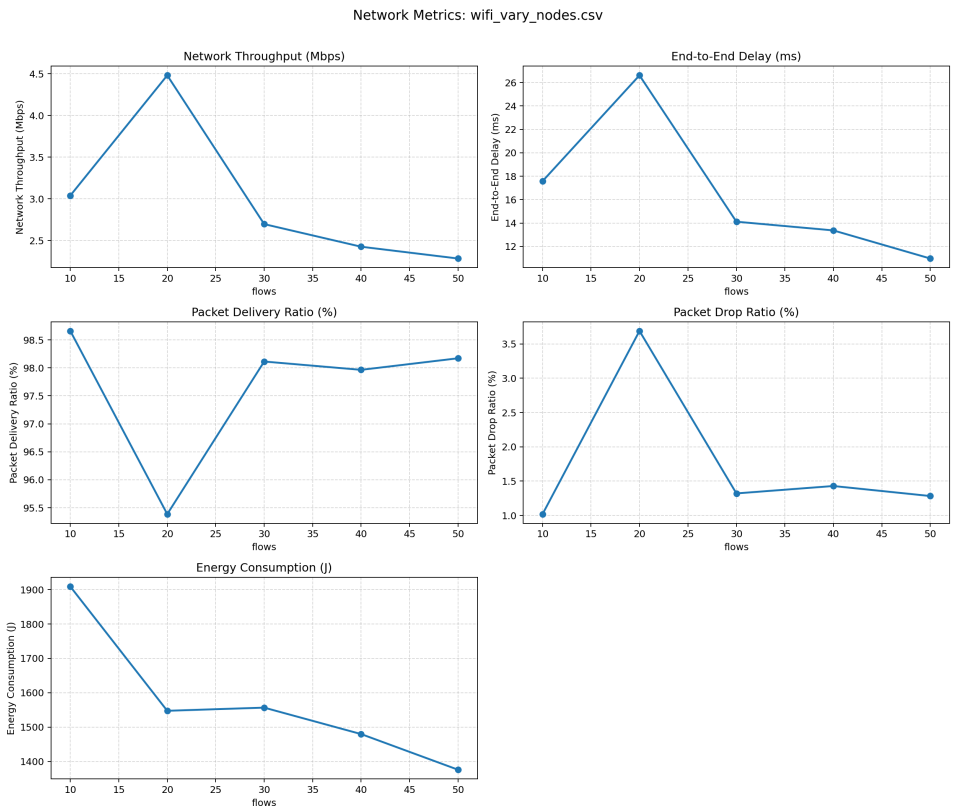

Figure 2: varied flows

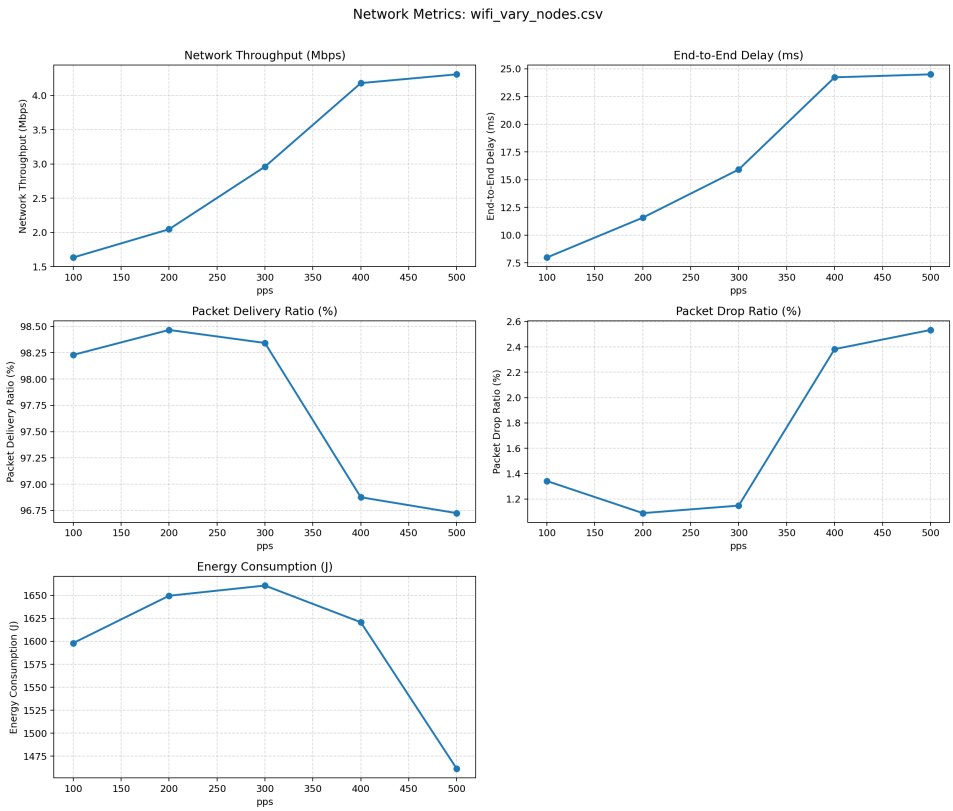

Figure 3: varied pps

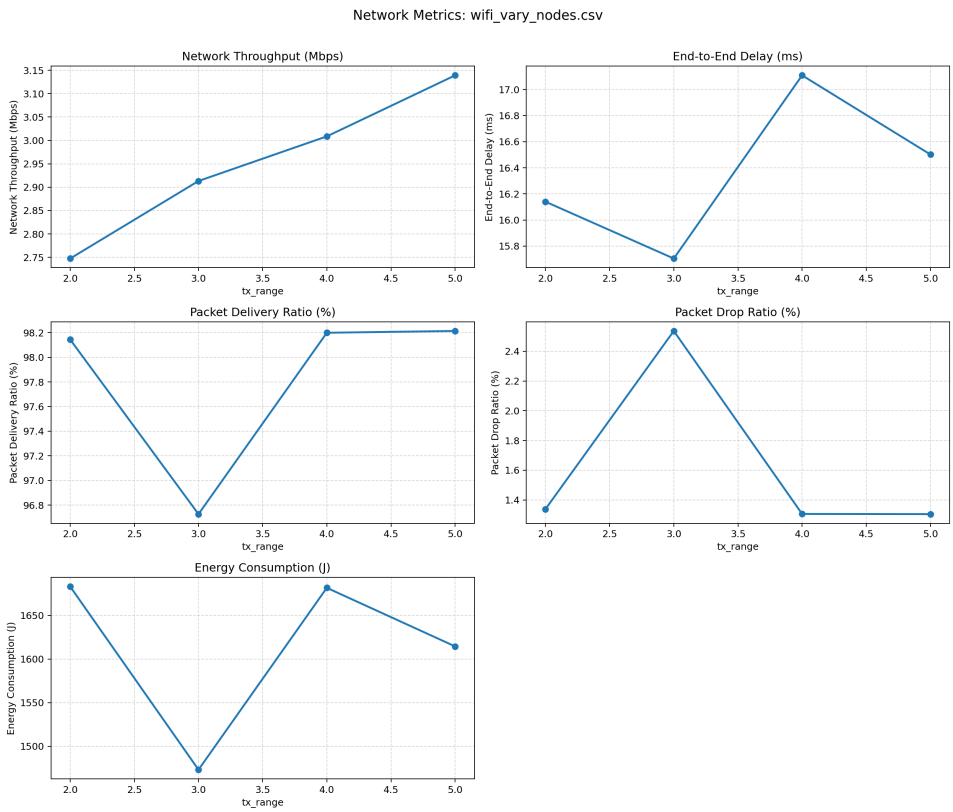

Figure 4: varied Rtx

8.2 for wifi-mobile-802.15.4

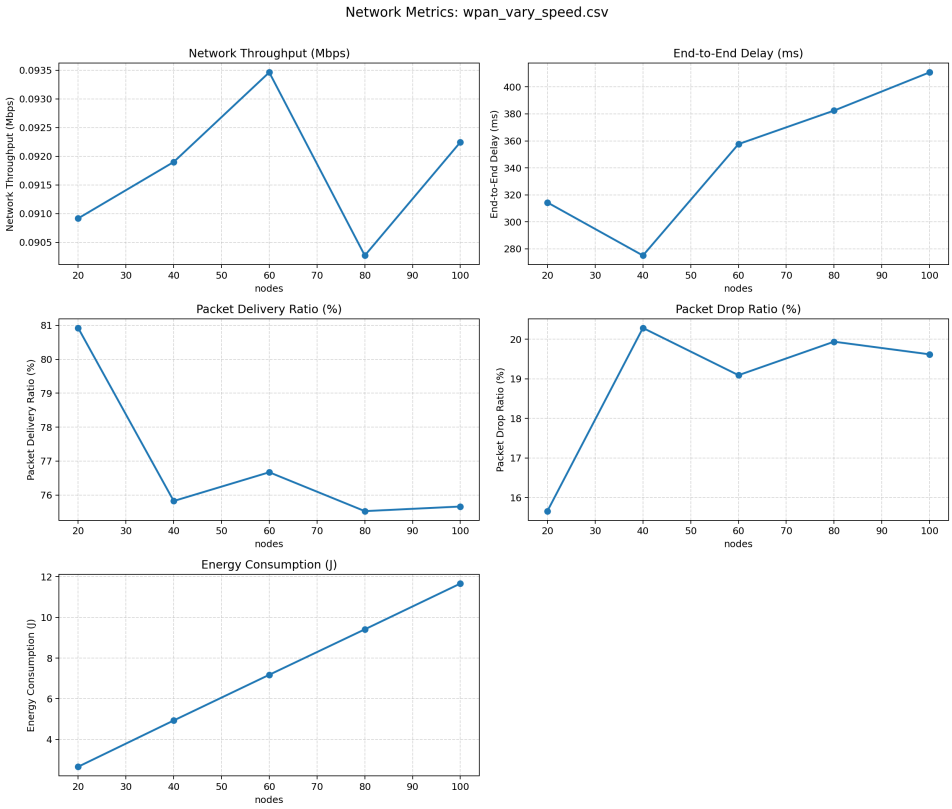

Figure 5: Varied nodes

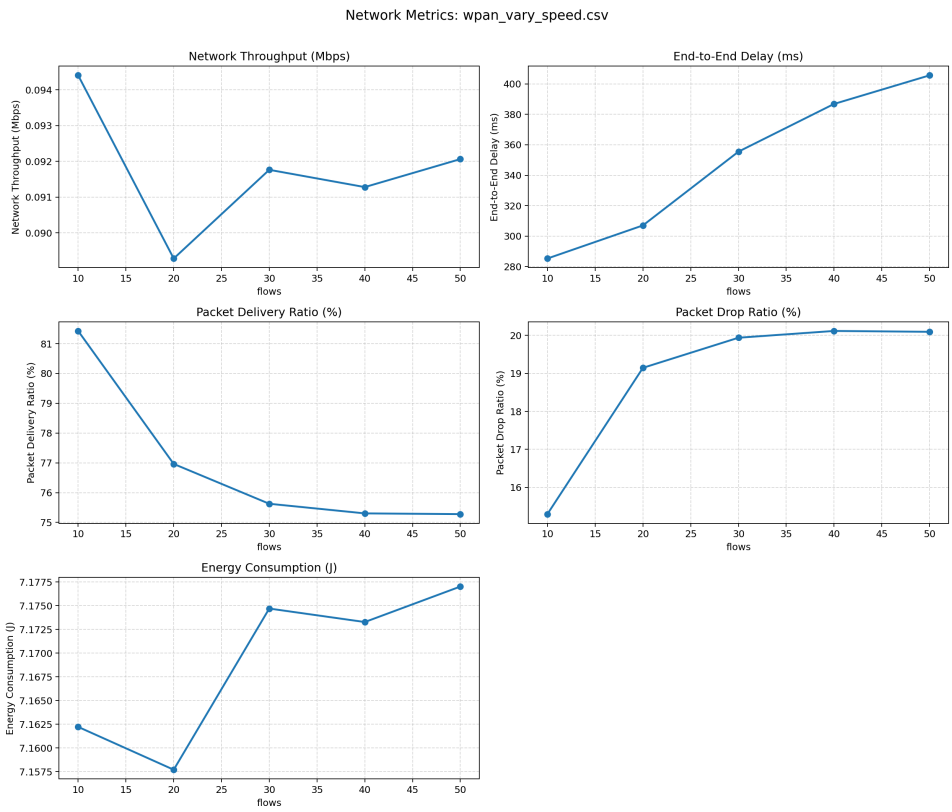

Figure 6: Varied flows

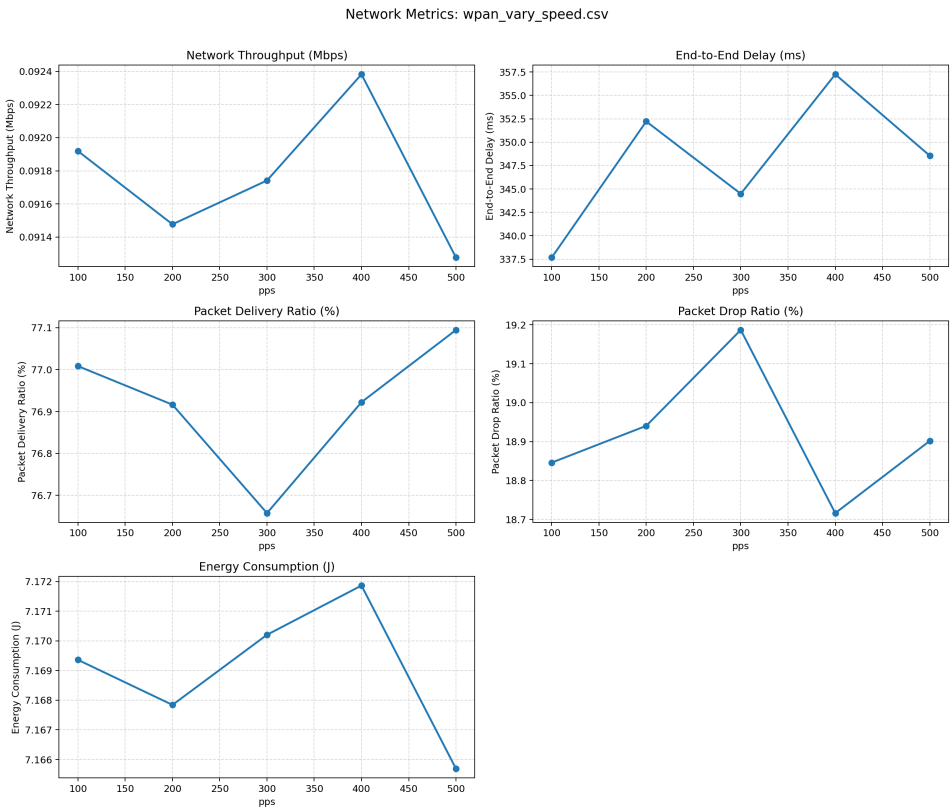

Figure 7: Varied pps

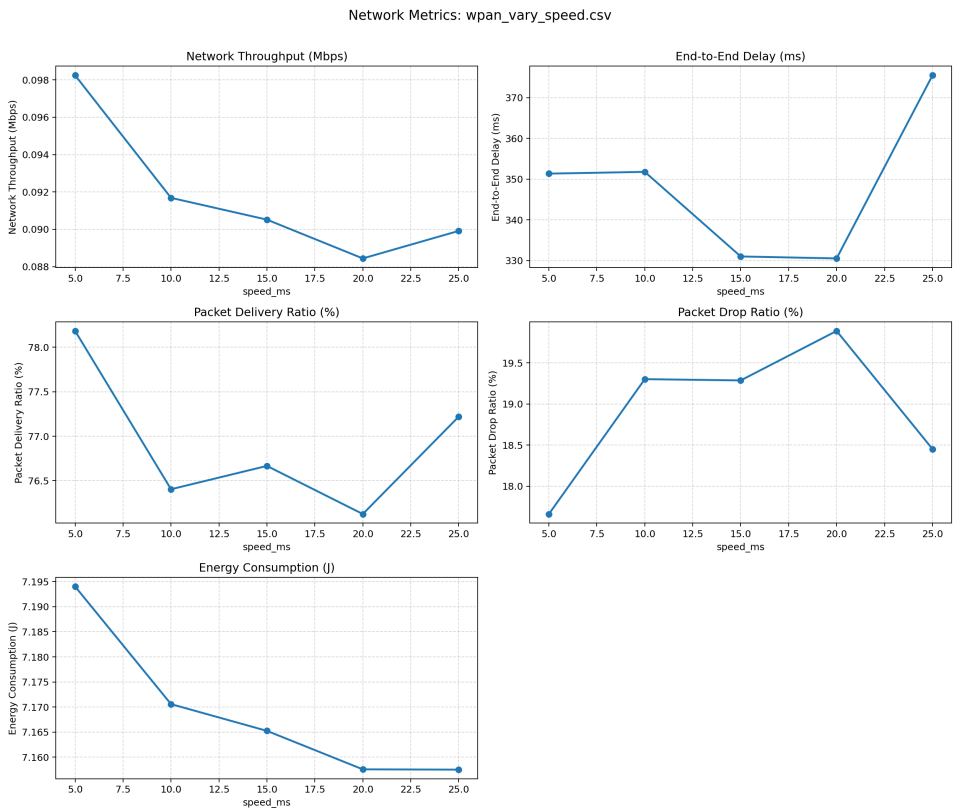

Figure 8: Varied spped

9 Summary Findings

- PDR is generally good (96–99.4) across tested scenarios, confirming that HyStart++ performs reasonably well in static 802.11g ad-hoc networks.
- Throughput degrades with increasing node density due to higher MAC contention and OLSR routing overhead — a known limitation of proactive routing in dense wireless setups.
- Energy consumption scales linearly with node count, dominated by idle listening and control traffic. In the WPAN topology, performance is stable but limited by the low inherent data rate.
- HyStart++ successfully completes its Conservative Slow Start phase and transitions smoothly, helping mitigate some burst losses compared to classic slow start. However, it cannot fully overcome MAC-layer interference and routing overhead in larger networks.
- Anomalies (e.g., PDR dip and delay spike at 80 nodes in WiFi) indicate a scalability threshold around 60–80 nodes under the current load and coverage.

10 Results

10.1 Wi-Fi (Static)

- Throughput decreases with nodes
- Delay shows V-shape (optimal at 60 nodes)
- PDR peaks at 60 nodes
- Energy increases linearly

10.2 802.15.4 (Mobile)

- Throughput slightly decreases
- Delay spikes at high speed (25 m/s)
- PDR around 78%
- Energy decreases with speed

11 Discussion

11.1 80-Node Anomaly

High interference causes PDR drop due to hidden terminal problem.

11.2 Energy Behavior

Energy decreases with speed due to reduced successful transmissions.

11.3 Routing Impact

OLSR convergence delay causes high latency at high mobility.

11.4 HYSTART++ Contribution:

HyStart++ (RFC 96) performed as designed by using delay signals to exit slow start early and applying a more conservative growth phase. This helped maintain relatively high PDR (mostly greater than 97) and reduced the severity of overshoot compared to standard NewReno. However, in wireless environments, packet losses are frequently caused by interference and MAC retransmission failures rather than pure buffer overflow. Therefore, transport-layer improvements like HyStart++ provide only partial mitigation — cross-layer solutions (better routing metrics, adaptive MAC parameters, or hybrid reactive/proactive routing) would likely yield bigger gains. The sharp performance degradation at 80 nodes suggests a critical density point where OLSR overhead and contention become dominant. This highlights the importance of testing beyond small-scale scenarios. The linear energy increase emphasizes the need for energy-aware protocols in battery-constrained ad-hoc deployments.

12 Conclusion

HyStart++ works well in Wi-Fi where congestion exists. In 802.15.4, losses are due to channel issues, so congestion control is ineffective.Overall, TcpHyStartPlusPlus is a suitable enhancement for TCP over static 802.11g ad-hoc networks, delivering good reliability under light-to-moderate traffic, but network-layer and MAC-layer bottlenecks remain the primary scalability limiters.